Answers learning task 13

2026 – 2027

Exercise 8.4: A confidence interval estimate is desired for the gain in a circuit on a semiconductor device. Assume that gain is normally distributed with standard deviation $\sigma = 25$.

(a) Find a $95\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(b) Find a $95\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(c) Find a $99\,\%$ CI for μ when $n = 10$ and $\bar{x} = 1000$.

(d) Find a $99\,\%$ CI for μ when $n = 25$ and $\bar{x} = 1000$.

(e) How does the length of the CIs computed above change with the changes in sample size and confidence level?

Answer

(a) $[984.5051, 1015.4949]$

(b) $[990.2002, 1009.7998]$

(c) $[979.6363, 1020.3637]$

(d) $[987.1209, 1012.8791]$

(e) Increasing the sample size n decreases the length of the CI, whereas increasing the confidence level increases the length of the CI.

Exercise 8.7: Consider the gain estimation problem in Exercise 8.4.

(a) How large must n be if the length of the $95\,\%$ CI is to be 30?

(b) How large must n be if the length of the $99\,\%$ CI is to be 30?

Answer

(a) $n \geq 11$

(b) $n \geq 19$

Exercise 8.9: Suppose that $n = 100$ random samples of water from a freshwater lake were taken and the calcium concentration (mg/L) measured. A $95\,\%$ CI on the mean calcium concentration is $0.52 \leq \mu \leq 0.74$.

(a) Would a $99\,\%$ CI calculated from the same sample data be longer or shorter?

(b) Consider the following statement: "There is a $95\,\%$ chance that μ is between 0.52 and 0.74." Is this statement correct? Explain your answer.

(c) Consider the following statement: "If $n = 100$ random samples of water from the lake

were taken and the $95\,\%$ CI on μ computed, and this process were repeated 1000 times, 950 of the CIs would contain the true value of μ." Is this statement correct? Explain your answer.

Answer

(a) Longer

(b) No, when the construction of a $95\,\%$-confidence interval for μ is repeated a large number of times, the true mean will belong to the calculated interval in $95\,\%$ of the cases.

(c) Yes, see explanation part (b).

Exercise 8.11: The yield of a chemical process is being studied. From previous experience, yield is known to be normally distributed and $\sigma = 3$. The past five days of plant operation have resulted in the following percent yields: 91.6, 87.34, 90.8, 89.65, and 91.3. Find a $95\,\%$ two-sided confidence interval on the true mean yield.

Answer

$[87.5084, 92.7676]$

Exercise 8.17: The life in hours of a 75-watt light bulb is known to be normally distributed with $\sigma = 20$ hours. Suppose you wanted the total width of the two-sided confidence interval on mean life to be six hours at $95\,\%$ confidence. What sample size should be used?

Answer

$n \geq 171$

Exercise 8.29: A postmix beverage machine is adjusted to release a certain amount of syrup into a chamber where it is mixed with carbonated water. A random sample of 25 beverages was found to have a mean syrup content of $\bar{x} = 30$ mL. Suppose the standard deviation is known and equal to $\sigma = 0.5$ mL. Find a $95\,\%$ CI on the mean volume of syrup dispensed.

Answer

$[29.8040, 30.1960]$

Exercise 7.63: Let $f(x) = (1/\theta)x^{(1-\theta)/\theta}$, $0 < x < 1$, and $0 < \theta < \infty$. Show that $\hat{\Theta} = -(1/n)\sum_{i=1}^{n} \ln(X_i)$ is the maximum likelihood estimator for θ and that $\hat{\Theta}$ is an unbiased estimator for θ.

Answer

The likelihood function is $L(\theta) = \theta^{-n} \prod_{i=1}^{n} x_i^{(1-\theta)/\theta}$. Taking the natural log gives:

$$\ln L(\theta) = -n \ln(\theta) + \frac{1-\theta}{\theta} \sum_{i=1}^{n} \ln(x_i)$$

Setting the derivative with respect to θ equal to zero:

$$\frac{d \ln L(\theta)}{d\theta} = -\frac{n}{\theta} - \frac{1}{\theta^2} \sum_{i=1}^{n} \ln(x_i) = 0 \implies \hat{\Theta} = -\frac{1}{n} \sum_{i=1}^{n} \ln(X_i)$$

To show it is unbiased, let $Y = -\ln(X)$. The CDF of Y for $y > 0$ is $P(-\ln(X) \leq y) = P(X \geq e^{-y}) = 1 - e^{-y/\theta}$, so $Y \sim$ Exponential(θ) with $E(Y) = \theta$. Thus:

$$E(\hat{\Theta}) = E\left(\frac{1}{n} \sum_{i=1}^{n} Y_i\right) = \frac{1}{n} \cdot n\theta = \theta$$

Hence, $\hat{\Theta}$ is an unbiased estimator for θ.